

In every game in this packet, two players take turns taking counters from a pile. Each problem says how many counters a player may take on a turn. Unless a problem says otherwise, whoever takes the last counter wins. If you cannot take any counters, you lose.

Problem 1: On each turn, a player takes 1 or 2 counters. For each starting pile in the table, decide whether you would rather go first or second. If you would rather go first, also write how many counters you take on your first turn.

Starting pile	First or second?	Counters you take first
5		
6		
7		
8		
9		
10		

Problem 2: Use the rule from Problem 1. Lee says, “With 21 counters, I will go second and always win. Whenever you take 1, I take 2, and whenever you take 2, I take 1.” Mo says, “With 20 counters, I will go first and take 1, and then I can always win.” Decide whether each claim is right. If a claim is right, explain why it always works. If it is wrong, show how to beat it.

Problem 3: Now on each turn a player takes 1, 3, or 4 counters. Jo starts with 10 counters and goes first. She takes 3, leaving 7, and says she can now win however her partner plays. Kai starts with 12 counters and goes first. He takes 4, leaving 8, and says the same thing. Decide whether each claim is right. If a claim is right, show how to answer every move the partner can make, all the way to the end of the game. If it is wrong, show how the partner can win.

Problem 4: Use the rule from Problem 3: take 1, 3, or 4 counters on each turn. For each starting pile from 1 to 20, decide whether you would rather go first or second. Shade the box of every pile where you would rather go second. In every other box, write how many counters you take on your first turn.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

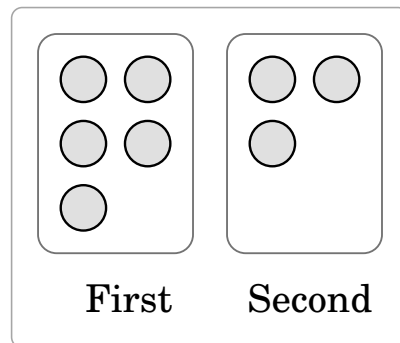
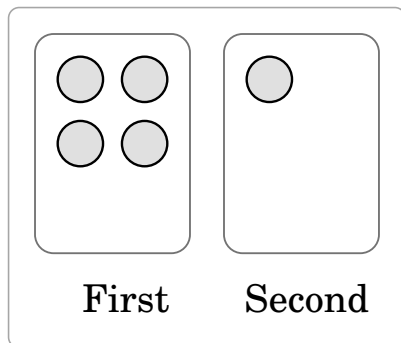
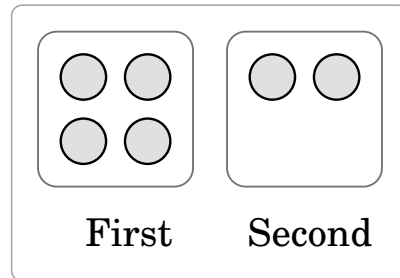
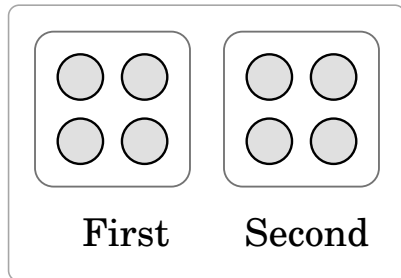
Problem 5: Keep the rule from Problem 3. For each starting pile in the table, decide whether you would rather go first or second. If you would rather go first, also write how many counters you take on your first turn.

Starting pile	First or second?	Counters you take first
30		
50		
61		
100		

Problem 6: Now on each turn a player takes 1, 3, or 5 counters. Shade the box of every starting pile from 1 to 20 where you would rather go second. Explain why the second player can always win from those piles.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Problem 7: Now there are two piles. On each turn, a player takes 1 or 2 counters from one of the piles. For each pair of piles below, circle whether you would rather go first or second. When the two piles are the same size, which player can always win? Explain why.



Problem 8: Find a rule for how many counters a player may take on a turn, so that you would rather go second exactly when the starting pile is 5, 10, 15, 20, 25, and so on. Then find a different rule that does the same thing.

Problem 9: Now whoever takes the last counter loses. On each turn, a player takes 1, 3, or 4 counters. Shade the box of every starting pile from 1 to 20 where you would rather go second.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20